

Isolation E_T Definition Comparison: 60 MeV Tower Threshold vs. 120 MeV vs. Topo-Cluster $R = 0.4$

PPG12 Automated Investigation

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1 Motivation

The isolated photon cross-section measurement relies on a reconstruction-level isolation energy cut to define the signal region in the ABCD method. Three isolation definitions are available in the slintree:

- **60 MeV tower threshold:** Sum tower E_T in $\Delta R < 0.3$ cone (EMCal + IHCAL + OHCAL), including only towers above 60 MeV. Branches: `cluster_iso_03_60_{emcal,hcalin,hcalout}`.
- **120 MeV tower threshold:** Same cone and calorimeters, but with a 120 MeV tower threshold. Branches: `cluster_iso_03_120_{emcal,hcalin,hcalout}`.
- **Topo-cluster $R = 0.4$:** Sum E_T of topological clusters in $\Delta R < 0.4$ cone. Branch: `cluster_iso_topo_04`. This is the **current nominal** definition (`use_topo_iso: 2` in the config).

This note quantifies the differences in run-by-run stability, ABCD yield, and signal/background discrimination power.

2 Signal/Background Discrimination (ROC Curves)

Using truth-matched signal (direct + fragmentation photons with truth $E_T^{\text{iso}} < 4 \text{ GeV}$ in $R = 0.3$) and background (jet-sample clusters) from the full MC simulation chain, we compute ROC curves for all seven available isolation definitions.

Table 1 summarizes the AUC values for the three primary definitions and all seven available.

Table 1: ROC AUC values by isolation definition and E_T bin.

Definition	10–15 GeV	15–20 GeV	20–25 GeV	25–30 GeV
70 MeV tower	0.820	0.849	0.857	0.861
60 MeV tower	0.806	0.847	0.855	0.859
120 MeV tower	0.805	0.847	0.857	0.857
topo $R = 0.3$	0.804	0.842	0.853	0.858
topo $R = 0.4$	0.823	0.871	0.875	0.878
topo soft $R = 0.3$	0.805	0.844	0.857	0.862
topo soft $R = 0.4$	0.827	0.873	0.881	0.883

Key findings:

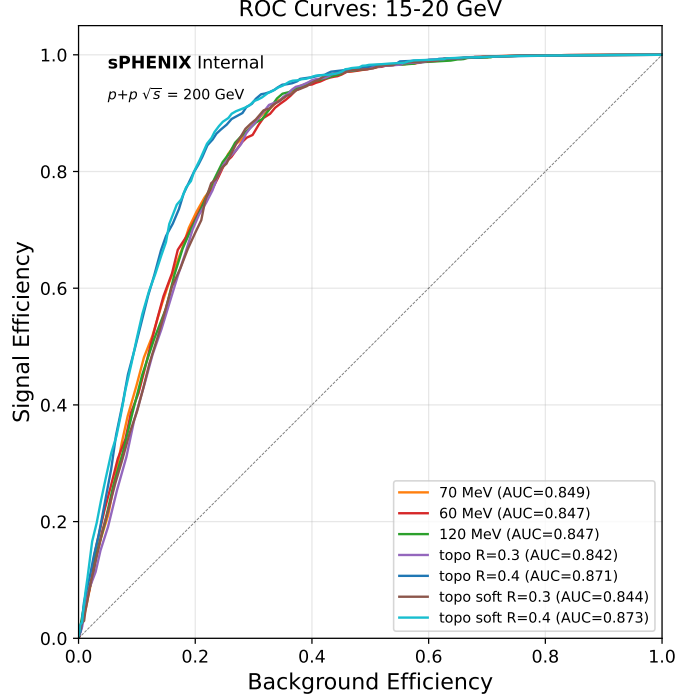


Figure 1: ROC curves for the 15–20 GeV E_T bin. The topo $R = 0.4$ definitions (blue/cyan) are clearly separated from the tower-threshold methods. AUC values are shown in the legend.

- Topo $R = 0.4$ outperforms tower-threshold methods by $\sim 2\%$ in AUC across all E_T bins. The advantage is largest at low E_T (10–15 GeV: 0.823 vs. 0.806).
- 60 MeV and 120 MeV tower thresholds are essentially identical ($<0.3\%$ AUC difference). The choice of tower threshold is negligible for discrimination.
- **The dominant factor is cone size, not method.** All $R = 0.3$ methods (tower 60/70/120 MeV, topo $R = 0.3$, topo soft $R = 0.3$) cluster together at $\text{AUC} \approx 0.80\text{--}0.86$, while both $R = 0.4$ methods stand $\approx 2\%$ above (Figure 3). The topo clustering algorithm itself provides minimal gain at fixed cone size; the larger cone captures more jet activity around background clusters, driving the improved rejection.
- Topo soft $R = 0.4$ is marginally better than standard topo $R = 0.4$ ($+0.2\text{--}0.5\%$ AUC).
- Among tower-threshold methods at $R = 0.3$, the 70 MeV threshold slightly outperforms 60 MeV at low E_T (0.82 vs. 0.81) but they converge at high E_T . 120 MeV is indistinguishable from 60 MeV.

3 Isolation E_T Distributions

The distribution shapes (Figure 4) explain the discrimination difference:

- **Signal:** Topo $R = 0.4$ has the sharpest peak near $E_T^{\text{iso}} = 0$, concentrating more signal at low isolation values.

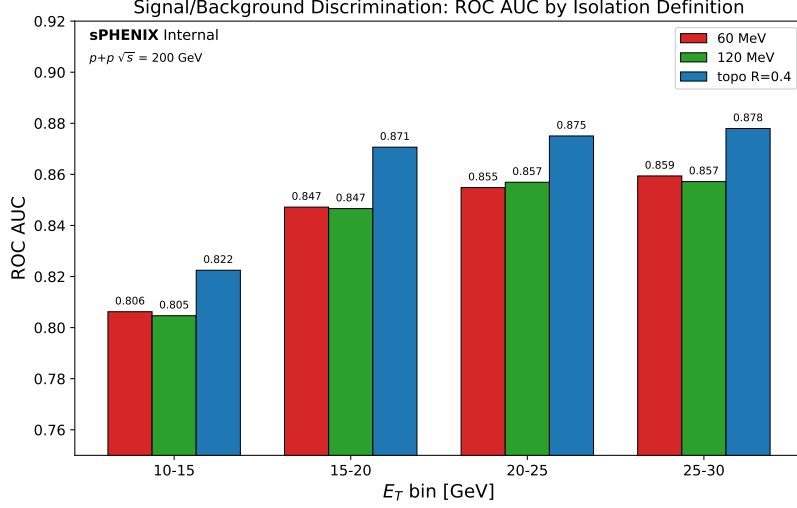


Figure 2: ROC AUC comparison across E_T bins for the three definitions of interest. Topo $R = 0.4$ consistently outperforms both tower-threshold methods by $\approx 2\%$ in AUC.

- **Background:** Tower-threshold methods produce smooth, broad distributions. Topo $R = 0.4$ has a distinctive spike near zero followed by a broader tail, yielding better separation.
- 120 MeV shifts both signal and background to lower E_T^{iso} compared to 60 MeV (fewer towers contribute), but the shift is similar for both categories, so discrimination is unchanged.

4 Run-by-Run Stability

Using the existing run-by-run QA infrastructure (`Cluster_rbr.C`), we compare the mean isolation E_T and ABCD region yields per run for all three definitions across all 1428 analyzed runs.

Table 2: Run-by-run stability summary (1428 runs).

Definition	$\langle E_T^{\text{iso}} \rangle$ [GeV]	RMS/Mean	Yield A [mb $^{-1}$]	RMS/Mean A	Yield C [mb $^{-1}$]
60 MeV tower	2.088	6.5%	6963	19.4%	10455
120 MeV tower	1.375	6.4%	10702	17.0%	17990
topo $R = 0.4$	1.993	4.6%	8606	17.4%	12148

Stability findings:

- Topo $R = 0.4$ has **30% less run-by-run variation** in mean E_T^{iso} (RMS/Mean 4.6% vs. 6.4–6.5% for tower methods).
- 120 MeV has a much lower mean isoET (1.38 vs. 2.09 GeV for 60 MeV) because the higher threshold removes towers between 60–120 MeV from the isolation sum.
- 120 MeV yields are **artificially inflated** (Yield A = 10702 vs. 6963 for 60 MeV) because the same parametric cut $E_T^{\text{iso}} < b + s \cdot E_T$ was optimized for topo $R = 0.4$, not re-tuned for the lower isoET scale. This does not reflect better physics performance.
- All three definitions show the same crossing-angle structure and outlier runs.

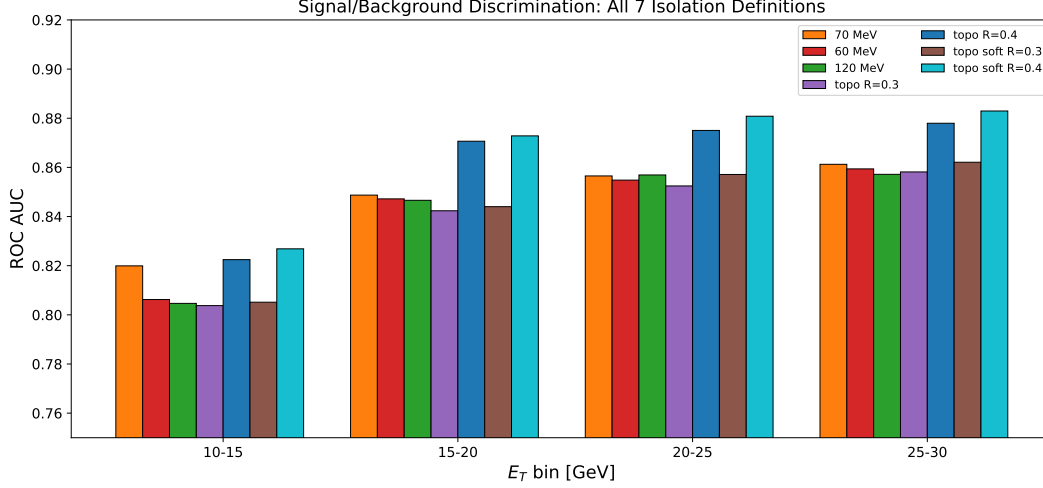


Figure 3: ROC AUC for all 7 isolation definitions. The hierarchy is dominated by cone size: all $R = 0.3$ methods cluster together, while $R = 0.4$ methods stand above.

4.1 ABCD Ratio Stability

The C/A ratio (Figure 7) is the key ABCD closure metric. All three definitions produce consistent ratios (~ 1.5 in 0 mrad, ~ 1.2 in 1.5 mrad), with the same outlier runs. This means the **ABCD background subtraction is insensitive to the isolation definition choice**.

5 Summary and Recommendation

Table 3: Overall comparison of the three isolation definitions.

Metric	60 MeV	120 MeV	topo $R = 0.4$
ROC AUC (15–20 GeV)	0.847	0.847	0.871
Mean isoET RMS/Mean	6.5%	6.4%	4.6%
Region A yield*	6963	10702	8606
C/A ratio stability	OK	OK	OK
Available in sim?	Yes	Yes	Partial

*Region A yields are not directly comparable across definitions because the same parametric cut was applied without re-optimization. The 120 MeV yield is inflated by the lower isoET scale.

Bottom line:

1. **Topo $R = 0.4$ is the best choice** for the nominal analysis: best discrimination (+2% AUC), best run-by-run stability (30% less variation), and a favorable yield/purity balance.
2. **60 MeV vs. 120 MeV is a wash:** $< 0.3\%$ difference in discrimination, essentially identical performance. The tower threshold choice has negligible physics impact.
3. **Cone size $R = 0.4$ is the key driver**, not the topo algorithm itself. All $R = 0.3$ methods (tower and topo) cluster together in AUC.

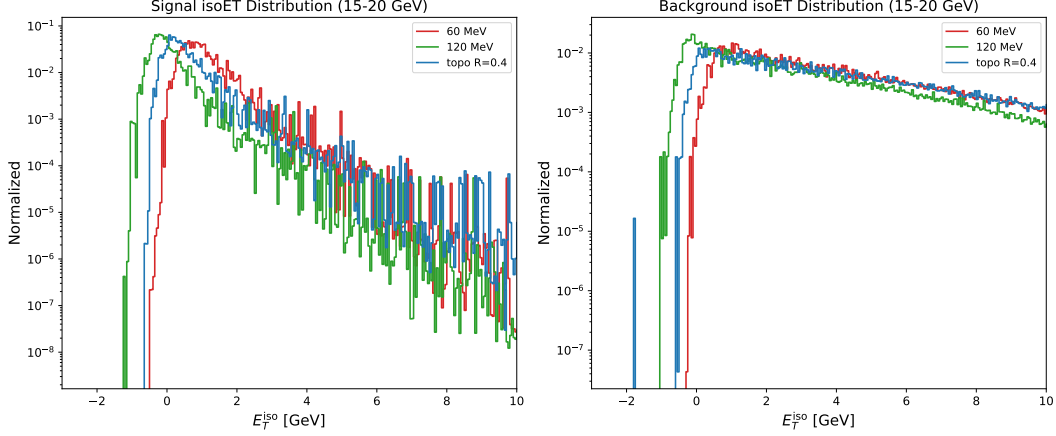


Figure 4: Normalized isolation E_T distributions for signal (left) and background (right) in the 15–20 GeV bin. Topo $R = 0.4$ produces a sharper signal peak near zero and a more spread-out background tail.

4. The current nominal config (`use_topo_iso: 2`) is **well justified**.
5. The 60 MeV tower threshold remains a valid cross-check definition for systematic studies (available in both data and sim).

6 Caveats (from Physics Review)

1. **Cone size conflation (WARN):** The topo $R = 0.4$ comparison mixes two effects: topo-cluster algorithm and larger cone size. The ROC analysis includes all 7 definitions (including topo $R = 0.3$) and shows the dominant factor is $R = 0.4$ vs. $R = 0.3$, not tower vs. topo. The run-by-run QA only compares three definitions and conflates these.
2. **Parametric cut not re-optimized (WARN):** The same $E_T^{\text{iso}} < b + s \cdot E_T$ cut (optimized for topo $R = 0.4$) is applied to all three definitions in the run-by-run ABCD. This makes yield comparisons unfair (especially 120 MeV, whose lower isoET scale inflates yields). The ROC comparison is unaffected (no cut applied).
3. **Missing intermediate jet samples (WARN):** The `IsoROC_calculator.C` uses jet5/12/20/30/40 but not jet8/10/15. The jet5 window (7–14 GeV) overextends into the jet8 domain, causing slight background misweighting at low E_T . Effect on ROC AUC is modest since curves are normalized.
4. **Config file (WARN):** `config_bdt_isoroc.yaml` does not exist; IsoROC was run with a manually provided config.

7 Code Changes

- `efficiencytool/Cluster_rbr.C`: Removed hardcoded `use_topo_iso = 0` override; added `iso_threshold == 2` for 120 MeV support; output filename now includes `var_type`.
- `efficiencytool/config_bdt_rbr_120MeV.yaml`: New config for 120 MeV run-by-run QA.

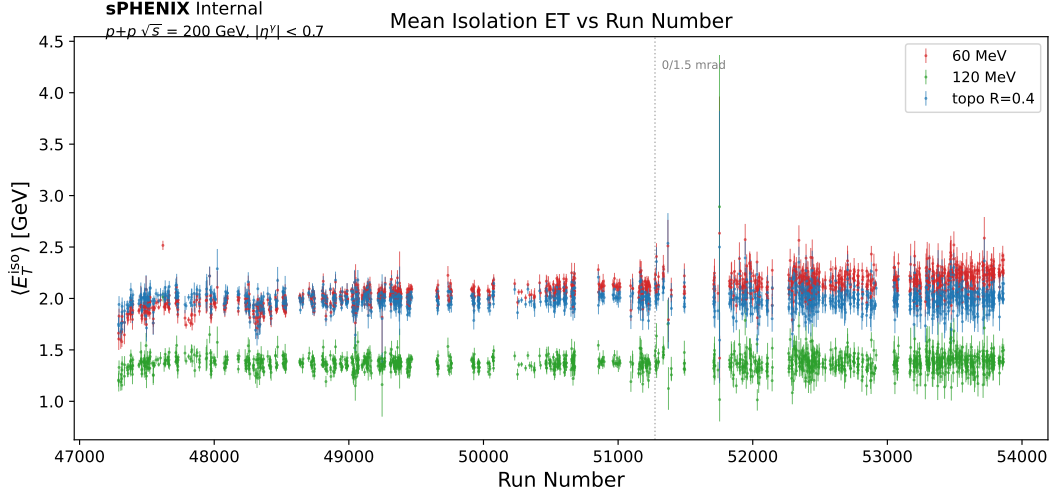


Figure 5: Mean isolation E_T vs. run number. Topo $R = 0.4$ (blue) runs ~ 0.1 GeV lower than 60 MeV (red) with visibly less scatter. The dashed line marks the 0/1.5 mrad crossing angle boundary.

- `plotting/plot_isoET_comparison.py`: New comparison plotting script (reads rbrQA files + ROC merged file, produces overlay plots and summary table).

8 Files Produced

- `figures/isoET_comparison/*.pdf` — comparison plots
- `efficiencytool/roc_plots/*.pdf` — per- E_T -bin ROC curves
- `reports/isolation_ET_comparison.pdf` — this report

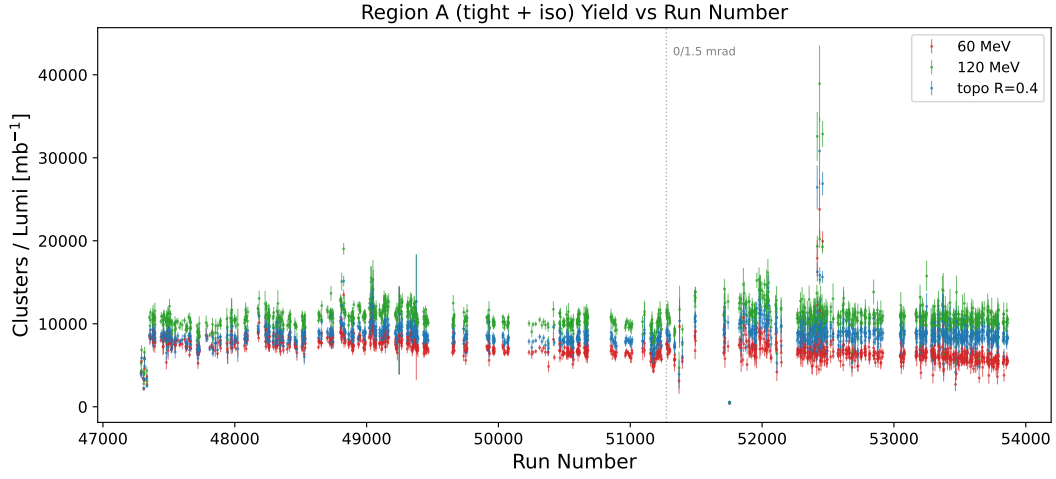


Figure 6: Region A (tight + isolated) cluster yield per lumi vs. run number. Topo $R = 0.4$ yields $\approx 24\%$ more signal-region clusters, consistent with the narrower isolation distribution.

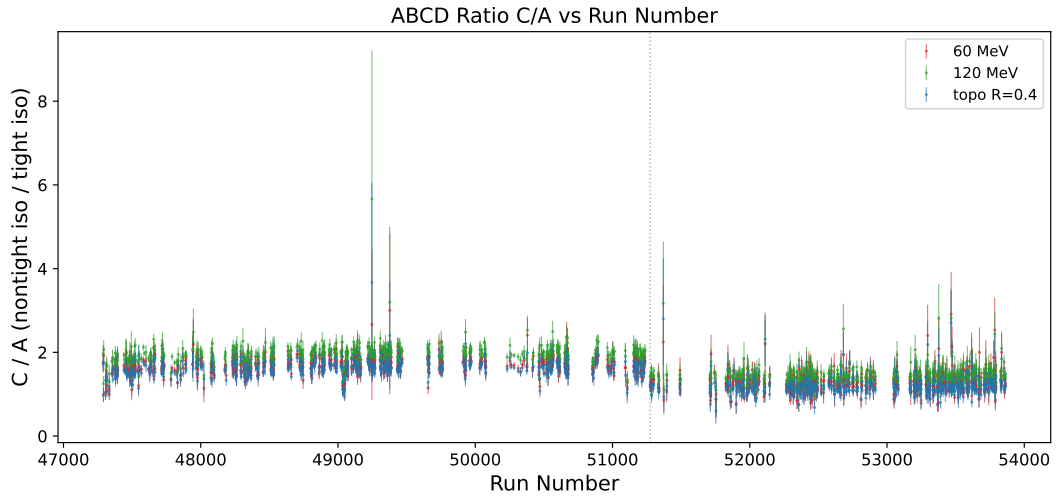


Figure 7: Ratio C/A (nontight-iso / tight-iso) vs. run number. All three definitions produce similar ratios, confirming that the ABCD closure is robust against the isolation definition choice.